EDGE AI OPTIMIZATION · PROJECT EVALUATION

Facial Recognition Optimization

Empirical assessment of post-training INT8 quantization on latency, biometric security, and robustness in a two-stage edge authentication pipeline.

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Target: Raspberry Pi 4 Model B

WHAT WE ARE TRYING TO DO

Motivation, Architecture, and Hypotheses

Dual-Stage Authentication

Real-world biometric security requires simultaneously matching enrolled identity while detecting photo, screen replay, or mask attacks on compute-constrained edge hardware.

System Pipeline

1. Face Detection & Alignment
Crops incoming webcam frame to required dimensions

2. Concurrent Inference Execution
Runs Liveness Detection and Identity Verification models

3. Binary Decision Gate
Accepts only when live status AND enrolled match pass

Core Hypotheses

H1 · EFFICIENCY
Post-training INT8 quantization will yield modest latency and active power reductions on the Raspberry Pi CPU.

H2 · ACCURACY
INT8 quantization will degrade the liveness model more than verification, since anti-spoofing depends on fine-grained texture cues rather than coarse embeddings.

H3 · FAIRNESS
Accuracy degradation will concentrate disproportionately under degraded inputs, such as low-light environments.

WHAT WE TRIED DOING

Experimental Methodology and Setup

MODELS

Pipeline Models

Face Detection:
InsightFace SCRFD-500MF (held fixed in FP32 across all tests as baseline preprocessor).

Liveness Detection:
Facenox MiniFASNetV2-SE (128×128 input), thresholded at the default cut off score: 0.0

Identity Verification:
InsightFace MobileFaceNet w600k_mbf (112×112 input), thresholded at 0.2421 cosine similarity.

MATRIX

Ablation Matrix

9 Total Configurations:
Full 3×3 grid pairing FP32, INT8 per-tensor, and INT8 per-channel across both models.

Quantization Engine:
ONNX Runtime 1.30 static PTQ, QDQ format, signed INT8 weights and activations.

Calibration Dataset:
MinMax calibration across 128 frames from p01 and p02 held-out live videos (seed 0).

SETUP

Hardware & Workloads

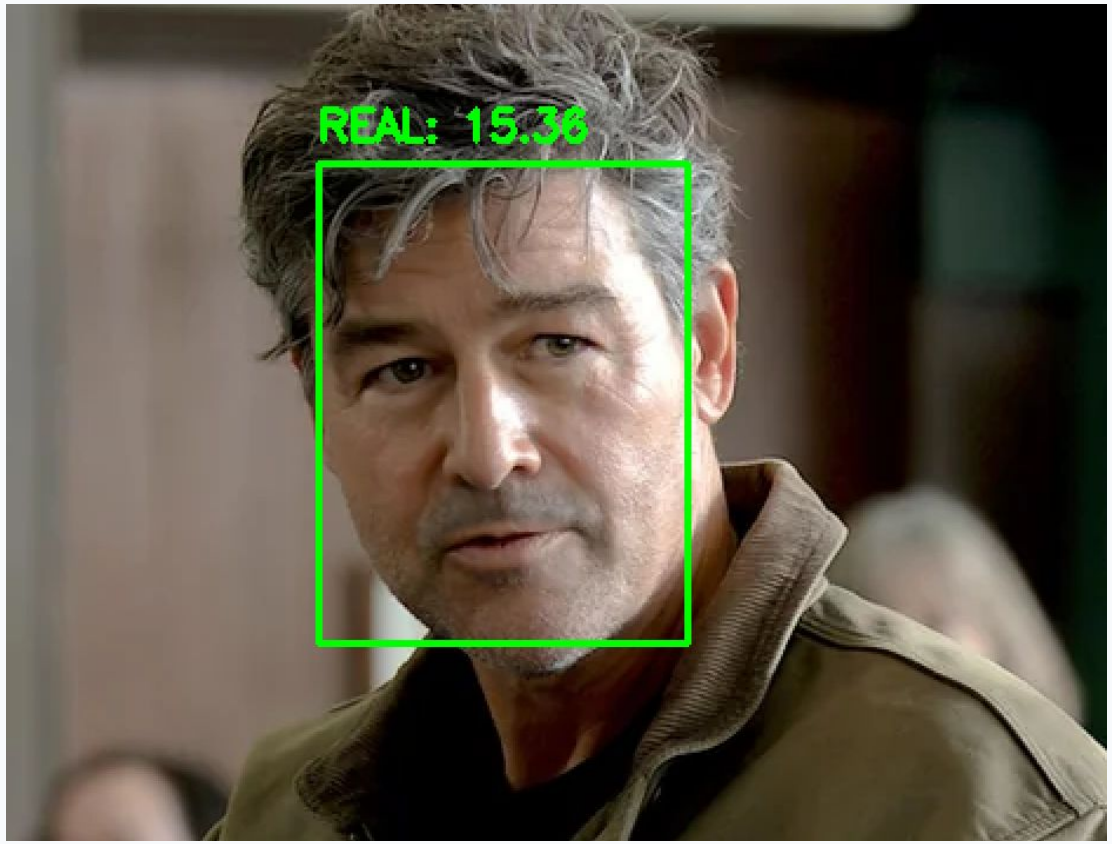
Target Rig:
Raspberry Pi 4 Model B (Cortex-A72 @ 1.8 GHz, 8 GB RAM) compared against x86-64 PC.

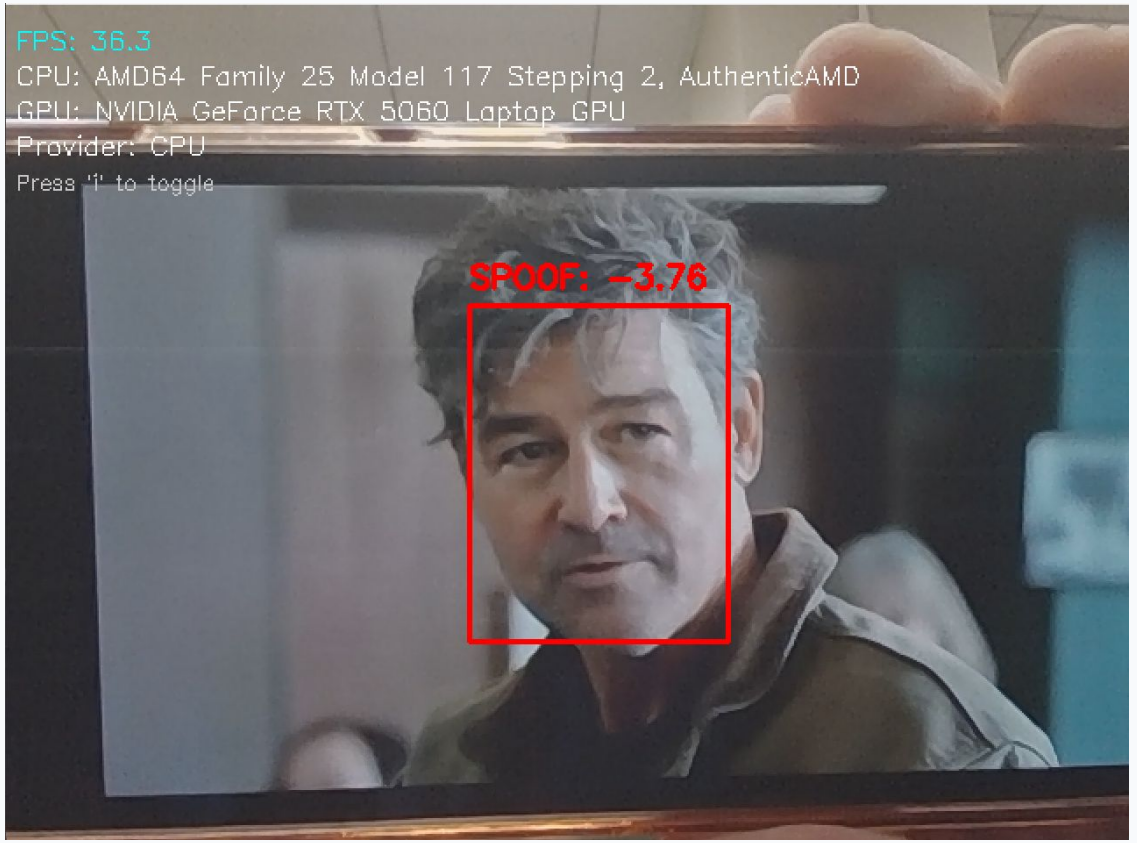
Datasets used:
LFW dataset[1] (5,965 photo pairs) for verification model, MSU-MFSD subset[2] (140 samples) liveness model, kaggle set[3] (406 frames) and 636 team video frames across 3 subjects for the combined pipeline.

1. https://www.kaggle.com/datasets/jessicali9530/lfw-dataset?resource=download
2. https://github.com/sunny3/MSU-MFSD
3. https://www.kaggle.com/datasets/tapakah68/anti-spoofing

Screenshots

Facenox (Anti-Spoofing/Liveness Test)





Screenshots

InsightFace (Facial Recognition)




```
Faces found: 6
Face 1: confidence=0.920
Embedding length: 512
Face 2: confidence=0.912
Embedding length: 512
Face 3: confidence=0.886
Embedding length: 512
Face 4: confidence=0.882
Embedding length: 512
Face 5: confidence=0.879
Embedding length: 512
Face 6: confidence=0.871
Embedding length: 512
Saved insightface_result.jpg

Process finished with exit code 0
```

WHAT THE RESULTS WERE

Key Results and Bottlenecks

Latency Inversion & Execution Time

+14% PIPELINE OVERHEAD

Full pipeline increased from **225.2 ms** (FP32) to **249–258 ms** (INT8) on Pi 4 (4 threads).

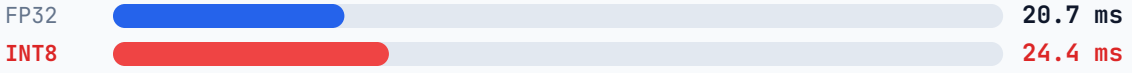
Identity Recognizer (Isolated)

+32% SLOWER



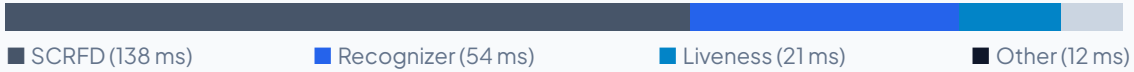
Liveness Model (Isolated)

+18% SLOWER



Frame Time Distribution (225.2 ms Total)

Detector Dominates (61%)



Biometric Accuracy Impact

Verification Parity (H2 Partial):

LFW EER unchanged at **0.40%–0.44%**; high embedding fidelity preserved.

Liveness Vulnerability Surge (H2 Confirmed):

Score shift (+0.75 to +1.5) increased attack acceptance on MSU-MFSD (**46.7% → 60–64%**)* and Kaggle (**16.2% → 22%**), while ranking (AUC 0.87) held.

*MSU's FP32 rate is high because the default cut-off sits at MSU's fake-face scores, not because the model can't tell them apart (AUC 0.99; about 3% of attacks would pass with a tuned cut-off). INT8's upward score shift pushes more fakes over that cut-off.

Identified Architectural Root Causes

Detector Overhead: FP32 SCRFD consumes 138 ms (61%), capping achievable quantization speedups.

PReLU Churn: ONNX Runtime lacks INT8 PReLU support, causing frequent cast conversions.

Depthwise & SDOT: Inefficient depthwise kernels; Cortex-A72 lacks vector INT8 dot-product instructions.

Baseline & Platform Caveats

Subject p03 Rejection: FP32 rejected p03 due to monitor glare and lighting, not quantization.

Memory Anomaly: Unexplained 2,000 MB vs 350 MB RAM allocation discrepancy on ARM runtime.

Power Proxy: Energy tracked via CPU time per frame; hardware power meter pending.

NEXT STEPS

Project Roadmap

Runtime & Model Adjustments

Selective Precision ("INT8 Except Depthwise")
Test selective quantization on the Pi to bypass depthwise INT8 conversion penalties and recover inference speed.

Face Detector Optimization
Address the 138 ms SCRFD-500MF bottleneck by quantizing the detector or evaluating downscaled input resolutions.

Decision Boundary Recalibration
Re-tune liveness threshold and calibration parameters to offset the +0.75 post-quantization score shift.

Hardware & Dataset Expansion

Hardware Power & Thermal Profiling
Deploy an inline USB-C power meter for direct wattage data

Memory Allocation Investigation
Diagnose the unexplained 2,000 MB vs 350 MB process memory reading on ARM to determine allocator behavior.

Multi-Seed & Dataset Expansion
Execute calibration seeds 1 and 2, record p03 without glasses to isolate glare, collect print attacks, and acquire official benchmarks.

Single Measurable System

Combine the Models
Build one webcam application that runs the face verification and liveness detection on each frame

Compare Execution Order
Test the applications running in parallel against liveness first, measuring latency, EER, liveness AUC, attack acceptance, etc.

Final Decision Rule
Accept access only when both the face is live and the identity match score exceeds the cut off